Code No: 115EM **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
SOFTWARE ENGINEERING
(Compute Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the advantages of using specialized process models?	[2]
b)	How does the increasing use of mobile devices impact software development and user expectations?	[3]
c)	What are the main challenges faced during requirements elicitation?	[2]
d)	Describe the classification of non-functional requirements?	[3]
e)	What is the role of maintainability in design quality?	[2]
f)	What are the typical steps involved in the interface design process?	[3]
g)	Explain black-box testing.	[2]
h)	What are the benefits of smoke testing?	[3]
i)	What are the objectives of risk monitoring?	[2]
j)	List and explain the objectives of formal technical reviews.	[3]

PART – B

(50 Marks)

2.a)	Give brief description about the waterfall model.	
b)	What are the advantages and disadvantages of replacing legacy software with new software?	[5+5]
	OR	
3.a)	What are the different maturity levels in the CMMI model, and how do they indicate an organization's process capability?	
b)	What are the potential security risks associated with using legacy software, and how can they be addressed?	[5+5]
4.a)	Describe the concept of security as a non-functional requirement and discuss the techniques and measures employed to ensure secure software development.	
b)	Describe the concept of inheritance in object models and its benefits in software design and reuse.	[5+5]
	OR	
5.a)	How are sequence diagrams used to model the interactions between objects or components in a software system?	
b)	Explain the process of gathering and eliciting user requirements.	[5+5]

6.a) How does usability testing contribute to user interface design?
b) Describe the layered architectural style and its benefits. [5+5]

OR

7.a) Discuss, how to reduce the user's memory load.
b) Describe the difference between qualitative and quantitative usability testing methods in interface design. [5+5]

8.a) Write and explain the depth first integration testing.
b) With the help of an example, explain the size oriented metrics. [5+5]

OR

9.a) Write in detail about the CK Metric suite.
b) How can we measure the quality of a software? Explain. [5+5]

10.a) How can we assess the risk impact? Discuss.
b) Discuss in detail about the sample-driven review process. [5+5]

OR

11.a) Describe the difference between risk components and risk drivers.
b) Give brief description on McCall's quality factors. [5+5]

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